Hopkins & Stryker Equation (HSE)

Todo comienza por inquietudes presentadas por el Profesor Jorge Petrosino respecto a la definición de la constante del recinto "R" utilizada por muchos autores para el calculo de la presión sonora generada por una fuente dentro de un recinto reverberante. La inquietud surge de la utilización de dicha constante en normativa y bibliografía de referencia en la acustica (como Beranek por ejemplo) calculada mediante la absorción de Sabine, mezclada con los cálculos de reverberación basados en los desarrollos de Eyring. Esta ensalada de superficies y coeficientes de absorción motivó una búsqueda de la raíz histórica en la definición de dicho parámetro.
La HSE en cuestión es la popularmente conocida formula:

$$L_p = L_w + 10\log_{10}\left(\frac{Q}{4\pi r^2} + \frac{4}{R}\right)$$

Mediante la búsqueda **acoustic "room constant"** en Google Scholar encontré la referencia a HSE en la siguiente publicación:

Jensen, Zachary R., "Improvements to the Two-Point In Situ Method for Measurement of the Room Constant and Sound Power in Semi-Reverberant Rooms" (2016). Theses and Dissertations. 5724. https://scholarsarchive.byu.edu/etd/5724
Abstract:
The two-point in situ method is a technique for measuring the room constant of a semi-reverberant room and the sound power of a source in that room simultaneously using two measurement positions. Using a reference directivity source, where the directivity factor along any given axis of the source has been measured, one is able to use the **Hopkins-Stryker** equation to measure both the room constant and the sound power level of another source rather simply. Using both numerical and experimental data, it was found that by using generalized energy density (GED) as a measurement quantity, the results were more accurate than those using squared pressure. The results also improved when one measurement position was near the source and the other measurement position was far from the source. This resulted in strong contributions of both the direct and reverberant fields in each of the measurement positions. Another improvement to the two-point method was the use of a local, spatial average around the measurement position. The assumptions in the Hopkins-Stryker equation rely on this average and it was found that a small local spatial average improved the measurements. However, this improvement was greater for squared pressure than for GED. Several source sound power levels and room constants were measured to show that these measurements are improved by using the suggested techniques.

Me llamo la atención desconocer de Hopskin & Stryker, la búsqueda me llevo a una presentación en PPT de **Leishmaniosis & Jensen** de 2014 en el contexto de la 167th Meeting

of the Acoustical Society of America titulada "A CLOSER LOOK AT THE HOPKINS-STRYKER EQUATION" en donde declaran los siguientes datos interesantes:
-The HSE (in original notation)

$$\rho_{av} = \frac{E}{4\pi c}\left[\frac{Q}{r^2} + \frac{16\pi}{R}\right]$$

was published by H. F. Hopkins and N. R. Stryker of Bell Labs in 1948 [Proc. I. R. E., 36, 315-335 (1948)].

-It provided the first concise description of the composite (direct plus reverberant) total energy density (TED) in a room.

-It was based on the Sabine-Franklin-Jaeger reverberant-field theory and closely tied to earlier work by V. O. Knudsen, H. F. Olson, F. Massa, E. Dietze, W. D. Goodale, and others.

-Beranek referenced their paper in his 1949 book Acoustic Measurements , but not in his better known 1954 book Acoustics, which provided useful HSE clarifications.

-As a result, some have mistakenly attributed the HSE to Beranek.

Con esta información me pareció razonable que no sea generalmente conocida como HSE y que la conozcamos por la popularización que logro Beranek de dicha ecuación.

Efectivamente en la publicación de Hopkins & Striker
Hopkins, H. F., & Stryker, N. R. (1948). A Proposed Loudness-Efficiency Rating for Loudspeakers and the Determination of System Power Requirements for Enclosures. Proceedings of the IRE, 36(3), 315–335. https://doi.org/10.1109/JRPROC.1948.226221

se define la constante del recinto:
The average energy density ρ_{av} at any point within the enclosure is the sum of the direct and reverberant energy density ρ_d and ρ_r.

$$\begin{aligned}\rho_{av} &= \frac{E \cdot Q}{4\pi r^2 c} + \frac{4E(1-\alpha)}{S_R c\alpha} \\ &= \frac{E}{4\pi c}\left[\frac{Q}{r^2} + \frac{16\pi}{R}\right] \quad (18)\end{aligned}$$

in which $\alpha S_R/(1-\alpha)$ is defined as the room coefficient R because of its flexible use in practical problems.

En la publicación, H&S mencionan a :
-V. 0. Knudsen, "Architectural Acoustics," John Wiley and Sons, New York, N. Y., 1932, p. 127.
-H. F. Olsen and F. Massa, "Applied Acoustics," P. Blakiston's Son and Co., Philadelphia, Pa.,second edition, 1939. (Olsen es Olson)

Knudsen es quien realiza el conjunto de mediciones con modelos a escala para determinar el valor del **Camino Libre Medio**

Olson y Massa desarrollan la base que toman H&S, En Applied Acoustics (1934) ya aparecia en la formula (1-α) / S α y hay referencias a Eyring (1920).

Los Popes que faltan meterse a la discusion eran Morse & Bolt (1944), no podian fallar. Leyendo su libro, me impacto una cita que realizan de
-Knudsen, V. O. (1934). Recent Developments in Architectural Acoustics. *Reviews of Modern Physics*, *6*(1), 1–22. https://doi.org/10.1103/RevModPhys.6.1

"The approximate theories, when used with caution and understanding, have served satisfactorily for the practical purposes of acoustical designing, and they will continue to do so until they are superseded by more exact theories."

Ademas dedican una sección de su capitulo sobre acústica geométrica de recintos a criticar la teoría:

9. Critique of Geometrical Theory

The widespread use of reverberation theory for the acoustical correction of auditoriums and lecture halls and the growing demand for acoustically designed buildings give ample evidence of the importance of the field so ably initiated by Sabine. Using reverberation theory with due consideration of a few simple stipulations which have been fully verified by wide experience, one can at present predict accurately the acoustic behavior of large auditoriums, music and speech rooms, theaters, and studios. Subject to a greater number of restrictions and empirically derived rules, one can also design small speech and music rooms, broadcasting studios, etc., though these cases require a degree of "good guessing" which is possible only after one has had considerable experience in following through the design, construction, and testing of rooms.

For the reduction of noise in rooms the reverberation theory is adequate for analyzing most cases, at least in large offices, industrial plants, and in any room in which the dimensions are fairly conventional: provided the absorbing material is distributed "properly".

There have been observed, however, numerous discrepancies between reverberation theory and measurements. Precalculated reverberation times are sometimes from 20 to 50 percent in error, regardless of the formula used, and in extreme cases, such as in very small rooms and at low frequencies, the discrepancy may be several hundred percent (H8, P9, S14, W10). In a room of complicated shape, such as an auditorium with a reverberant back-stage, or a large rotunda with a dome ceiling, the reverberation may consist of two portions having widely different decay rates. The reverberation in coupled spaces has been studied (D1), but it is not fully understood.

(H8) F. U. Hunt, "Absorption coeficient problem",J.Acous. Soc. Am. 11, 38 (1939).
(P9)J. R. Power, "Measurement of absorption in rooms with sound absorbing ceilings",J. Acous.Soc. Am. 10, 98 (1938)
(S14) G. T. Stanton, "Correlation of sound absorption coefficients with field measurements", J. Acous. Soc. Am. 11, 45 {1939)
(W10) F. J. Willig, "Comparison of sound absorption coefficients obtained experimentally by different methods", J. Acous. Soc. Am. 10, 293 (1939
(D1)A. H. Davis, "Reverberation equations for two adjacent rooms connected by incompletely sound-proof partition",Phil. Mag. 50, 75— 80 (1925)

Most of the difficulties in applying reverberation theory to rooms arise when the simple geometric assumptions are not fulfilled. **Sound is often not random in its propagation nor is it uniformly diffuse throughout the room.** As a result, in these cases, the decay is not logarithmic, is modulated by small irregularities, has major breaks in the curve, and is not the same everywhere in the room. Even more serious from the geometrical point of view is the observation that in many cases the effective absorption of an absorbing material is not independent of area, nor of position in the room. Extensive measurements have been made on this "area effect" and "pattern effect" (A2, C2, D4, E4, P7, R2, S3), but a calculation of these effects has had to await the development of wave acoustics.

(A2) Andree, C. A. (1932). Effect of position on the absorption of materials for the case of a cubical room. *Journal of the Acoustical Society of America*, 3, 535.
(C2) Chrisler, V. L. (1934). Sound absorption coefficients. *Journal of the Acoustical Society of America*, 6, 115.
(D4) Dreisen, I. (1936). The orientation of natural acoustic vibrations in a room with concentrated absorbents and of anomalous shapes. *Technical Physics of the USSR*, 3, 743.
(E4) Eyring, C. F. (1930). Conditions under which residual sound in reverberant rooms may have more than one rate of decay. *Journal of the Society of Motion Picture Engineers*, 15, 528. *(Nota: El texto original decía 1943, pero el Vol. 15 corresponde a 1930, que es la fecha correcta de este artículo clásico).*
(P7) Parkinson, J. S. (1930). Area and pattern effects in the measurement of sound absorption. *Journal of the Acoustical Society of America*, 2, 112.
(R2) Ramer, L. G. (1941). Absorption of strips, effects of width and location. *Journal of the Acoustical Society of America*, 12, 323.
(S3) Sabine, P. E. (1922). Diffraction effects in sound absorption measurements. *Physical Review*, 19, 402.

Probably the most serious defect of the geometrical approach has been its inability to yield consistent, reliable, standard measurements of absorption coefficients of materials. In a recent symposium on this problem, Hunt discussed the "absorption coefficient problem" (H8) and summarized the replies to questionnaires sent to a large number of groups of acousticians with the following statements: (1) Coefficients of the same material measured in different laboratories are not usually in agreement. (2) Field measurements yield smaller coefficients than laboratory measurements. (3) Increasing the sample size leads to smaller coefficients, but this effect will not explain the preceding discrepancies.

In the same symposium P. E. Sabine (S7), C. F. Eyring (E5), and others (S14) discussed the absorption measurement problem from various viewpoints. Provided standardized test procedures, arbitrarily specified, are followed by different laboratories, the variations in measured coefficients may be reduced considerably, particularly at higher frequencies. But even with due precautions, different laboratories continue to yield coefficients which in some cases differ as much as 20 percent.

(S7) Sabine, P. E. (1939). Architectural acoustics, its past and its future. *Journal of the Acoustical Society of America*, 11, 21.
(E5) Eyring, C. F. (1939). Symposium discussion. *Journal of the Acoustical Society of America*, 11, 104.
(S14) Stanton, G. T. (1939). Correlation of sound absorption coefficients with field measurements. *Journal of the Acoustical Society of America*, 11, 45.

Increíble que 100 años mas tarde sigamos teniendo el mismo problema para medir absorción, que las diferencias entre laboratorios sigan siendo tan significativas como en 1930, y que sigamos teniendo las mismas discusiones sobre "difusividad" del campo sonoro, area de la muestra, etc.

Es interesante ver el texto de Beranek donde si hace referencia a Hopkins & Stryker, y es curioso que nunca mas en sus libros haya referenciado a dicho autores.

Rescato estas conclusiones de la presentación de **Leishmaniosis & Jensen** :

-The Hopkins-Stryker is an important historic equation of room acoustics.

-It could be better implemented if users were more familiar with its origins, assumptions, and potential applications.

-The locally averaged energy density, sound power, directivity factor, distance from the acoustic center, and room constant are all key elements of the equation that merit special attention.